Astha Gupta | MA25M102

M.Sc. in MATHEMATICS AND STATISTICS · Indian Institute of Technology, Tirupati
Email: ma25m102@iittp.ac.in | Phone: 6206361067 | LinkedIn: LinkedIn | GitHub: GitHub

EDUCATION

YEAR	DEGREE / EXAM	INSTITUTE	CGPA / MARKS
2025-2027	M.Sc. (Mathematics)	Indian Institute of Technology, Tirupati	**7.70**
2024	B.Sc. (Science)	IGNOU	**63.36%**
2020	Senior Secondary	CBSE	**71.6%**
2018	Secondary	CBSE	**91.8%**

PROJECTS

Physics-Informed Neural Networks for Additive Manufacturing | Indian Institute of Technology Tirupati — **May '26 - Present**

Supervisor: Prof. Dr. Panchatcharan, Dr. Kiran · Tools & Technologies: Python | PyTorch | PINNs | Fourier Neural Networks | Deep Learning | Numerical Methods

- **Developed** Physics-Informed Neural Networks (PINNs) and Fourier Neural Networks for thermal modeling in additive manufacturing
- **Implemented** GradNorm, normalized losses and adaptive weighting for improved convergence
- **Performed** single-GPU and dual-GPU training to evaluate scalability and computational efficiency
- **Built** the framework using PyTorch and scientific computing libraries

Physics-Informed Neural Networks for Navier-Stokes Equations — **May '26 - Present**

Supervisor: Prof. Dr. Panchatcharan · Tools & Technologies: Python | PyTorch | PINNs | FDM

- **Developed** PINNs for 2D and 3d incompressible Navier-Stokes lid-driven cavity flow , Poisson flow , Couette flow
- **Investigated** adaptive weighting, optimizer scheduling and activation functions
- **Benchmarked** PINN predictions against Finite Difference Method solutions and Ghia benchmark
- **Compared** single- and dual-GPU implementations

HireMate - AI Resume Analyzer | Github — **Jun '26**

Tools & Technologies: React | FastAPI | Python | SQLite | Git

- **Developed** a full-stack resume analyzer and job matching platform
- **Built** REST APIs for authentication and resume management
- **Designed** scalable backend architecture for AI/LLM integration
- **Worked** Big Data transformation using PySpark script when it ingests thousands of job postings from raw web scrapers

Image and Video Compression using Tensor Decomposition

Tools & Technologies: Python , Scipy , Numerical Methods

- **Implemented** Tucker and CP tensor decomposition algorithms for multidimensional image and video compression.
- **Achieved** significant storage reduction while preserving reconstruction quality measured .
- **Compared** tensor-based compression with conventional matrix decomposition methods (SVD).

Robotic and Drone Trajectory Planning using Cubic Numerical Interpolation

Tools & Technologies: Python , Numerical Methods , Scipy

- **Implemented** Hermite cubic numerical interpolation to calculate trajectory coordinates (x, y, z)
- **Extracted** geographical datasets from Google Maps to generate a continuous path

TECHNICAL SKILLS

Programming	Python | C++ | SQL| R | Matlab
Data Engineering and Libraries	NumPy | Pandas (Data Manipulation) | Data Curation and Visualization | PySpark | SciPy | SymPy | Matplotlib | Scientific Computing | Mathematical Modeling
Machine Learning & Deep Learning	PyTorch | Autograd | GradNorm | Neural Networks (PINNs and FNNs) and Operators | Optimizer Scheduling
Frameworks	FastAPI | React
Developer Tools & Workflow	Git | GitHub | Linux | VS Code | PyCharm | Google Colab | Jupyter Notebook

KEY COURSES TAKEN

Data Curation and Visualization | Scientific Computing | Mathematical Modelling | Numerical Analysis | Linear Algebra | Statistical Inference | Data Science Laboratory | Basic Programming Laboratory

POSITIONS OF RESPONSIBILITY

Placement Representative (PR) | M.Sc. Mathematics and Statistics, IIT Tirupati — **Present**

Class Representative (CR) | M.Sc. Mathematics and Statistics, IIT Tirupati — **Jul '25 - Present**

Social Media Head (Core Member) | The Statistics Club — **Present**

Self declared by the student. Career Development Cell, IIT Tirupati could not verify the relevant documents.

ACHIEVEMENTS

- **Solved** 150+ LeetCode problems covering Data Structures and Algorithms.
- **Awarded** institute scholarship based on academic performance.

CERTIFICATIONS

- **Introduction** to Machine Learning — Kaggle (2026)
- **Intermediate** Machine Learning — Kaggle (2026)
- **Intro** to Deep Learning — Kaggle (2026)
- **MATLAB** Onramp | Statistics Onramp | Machine Learning Onramp — MathWorks (2025)
- **How** MATLAB Graphics Work | Intro to Linear Algebra with MATLAB — MathWorks (2025)

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